Introduction to Computer Science
Fall 2026
Homework 8

Due: November 22, 2026 at 23:58:59
No Copy-and-Paste. No Late Submission.

Instructions. Please write your answers to each problem using LaTeX and submit the compiled PDF file. If you find a problem difficult to understand or solve, you may use external resources, including search engines (such as Google), online references (such as Wikipedia), and AI tools (such as DeepSeek), to study relevant concepts and assist your learning. However, you are responsible for understanding your submitted solution, and **please make sure every word in your submission is written by yourself to avoid plagiarism**. If you use any AI tool while working on the assignment, please include a short disclosure at the end of your submission stating which tool you used and briefly describing how you used it. Undisclosed use of AI tools may be subject to point deductions.

1 Randomized Patrols and Strategic Responses (5 points)

Consider a single-round patrol problem with three sites, A, B, and C. One patrol team protects exactly one site, and an attacker attacks exactly one site. An attack on the protected site causes zero loss. An attack on an unprotected site causes a loss of 1, 2, or 3 units at A, B, or C, respectively. There are no other costs.

A **pure strategy** always protects the same site. A **mixed strategy** chooses a site according to a probability distribution $\mathbf{p} = (p_A, p_B, p_C)$, where

$$p_A, p_B, p_C \geq 0, \qquad p_A + p_B + p_C = 1.$$

Zero probabilities are allowed, so pure strategies are special cases of mixed strategies. The attacker knows $\mathbf{p}$ but must choose a target **before observing which site is actually protected**. The attacker chooses a target that maximizes the expected loss, meaning the probability-weighted average of the possible losses.

Let $L(\mathbf{p})$ denote the **worst-case expected loss** under this timing, when the attacker chooses a target to maximize the expected loss. The defender wants to minimize $L(\mathbf{p})$. Give exact values and justify your answers.

(a) Derive an expression for $L(\mathbf{p})$ in terms of p_A, p_B, p_C. Find the worst-case loss of each pure strategy and identify the best one. Then compute $L(\mathbf{p})$ for the uniform strategy $\mathbf{p} = (1/3, 1/3, 1/3)$. Does uniform randomization improve on the best pure strategy?

(b) First restrict attention to strategies with $p_A = 0$. Among these strategies, find the probabilities p_B, p_C that minimize $L(\mathbf{p})$, and give the minimum value. Briefly justify your choice.

(c) Now remove the restriction $p_A = 0$. Is your strategy from (b) still optimal among all mixed strategies? Prove your answer.

(d) Change only the timing: the protected site is now sampled from $\mathbf{p}$ and revealed to the attacker **before** the attacker chooses a target. Derive the expected loss under this new timing, then find the optimal patrol distribution and its loss. Explain why the conclusion differs from (c).

2 Evaluating AI Systems for Social Good (5 points)

In Prof. Fei Fang's lecture, we learned how AI can support environmental conservation by automatically identifying relevant news articles. Inspired by NewsPanda and more recent LLM-based approaches, suppose a non-governmental organization (NGO) considers two AI systems:

- **System A (Fine-tuned BERT Classifier):** A BERT-based classifier fine-tuned on 2,000 labeled news articles.
- **System B (Few-shot LLM Classifier):** A pretrained LLM that performs classification using 10 labeled examples in its prompt, without additional fine-tuning.

Both systems classify each article as either *relevant* or *irrelevant* to environmental conservation. All numerical values below are hypothetical.

Part I: Classification Performance

Both systems are evaluated on the same independent test set of 2,000 articles, of which 200 are relevant and 1,800 are irrelevant.

The evaluation produces the following results:

	System A	System B
Articles predicted as relevant	170	230
Correctly predicted relevant articles	150	180

For each system, calculate its accuracy, precision, recall, and F1-score. Report your answers as percentages rounded to two decimal places. Hint: you may need to compute the true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) first, and then use the following definitions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Part II: Computational Cost

The NGO now considers the computational cost of deploying the systems.

	System A	System B
One-time task-specific training cost	6,300 units	0 units
Inference cost per article	0.05 units	0.35 units

In the following computation, exclude the original pretraining and the annotation costs for either model. The per-article inference cost 0.35 for System B already includes processing the 10 labeled few-shot examples. All costs are expressed in the same normalized units. Let N be the number of articles to be processed. Express the total computational cost of each system as a function of N. Find the number of articles at which the two systems have equal total computational cost. Determine which system is computationally cheaper when N is larger than this number.

Part III: Considering the Cost of Errors

In environmental conservation, missing an important news article can delay action, while incorrectly flagging an irrelevant article wastes human review time.

Suppose the NGO assigns the following penalties:

- Each **false negative** incurs a penalty of 8 cost units.
- Each **false positive** incurs a penalty of 2 cost units.

Define the overall cost as

$$C_{\text{overall}} = C_{\text{training}} + C_{\text{inference}} + C_{\text{errors}}.$$

Assume that the test-set distribution and classification error rates remain unchanged. All error penalties use the same normalized cost units as the computational costs. Derive the expected overall cost of each system as a function of N. Find the number of processed articles at which their expected overall costs are equal.

Remark Let N denote the number of new articles processed after evaluation. For the 2,000 test articles discussed in Part I, exclude the cost of evaluating them from all cost calculations in Part II and III.

Declaration of AI Use

Please disclose any use of generative AI tools in completing this homework. For each tool used, briefly state what you used it for. You remain responsible for the correctness and originality of your submission. If you did not use any generative AI tools, remove this section.